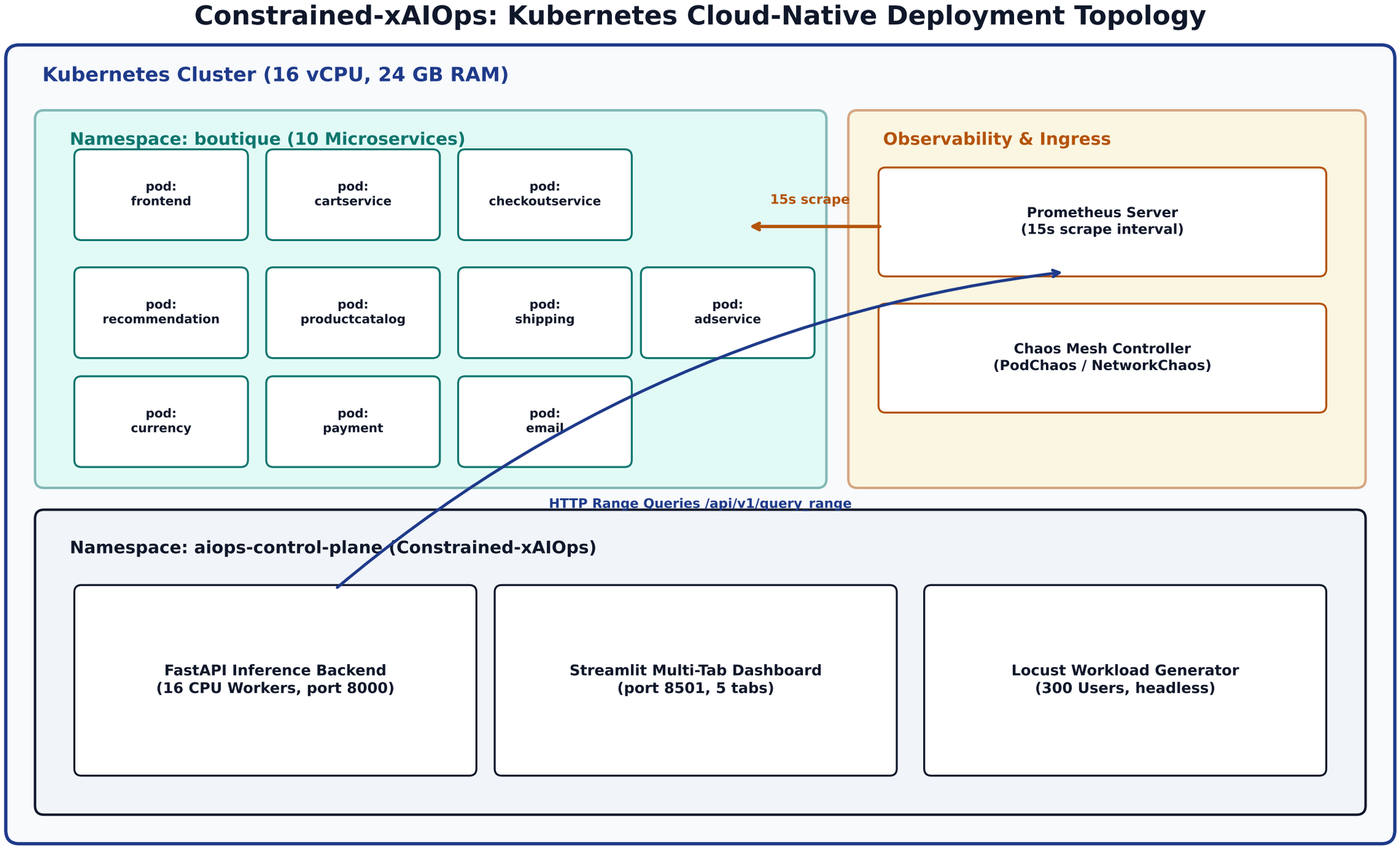
Constrained-xAIOps: Kubernetes Cloud-Native Deployment Topology
Kubernetes Cluster (16 vCPU, 24 GB RAM)
Namespace: boutique (10 Microservices)
pod: frontend
pod: cartservice
pod: checkoutservice
pod: recommendation
pod: productcatalog
pod: shipping
pod: adservice
pod: currency
pod: payment
pod: email
15s scrape
Observability & Ingress
Prometheus Server (15s scrape interval)
Chaos Mesh Controller (PodChaos / NetworkChaos)
HTTP Range Queries /api/v1/query_range
Namespace: aiops-control-plane (Constrained-xAIOps)
FastAPI Inference Backend (16 CPU Workers, port 8000)
Streamlit Multi-Tab Dashboard (port 8501, 5 tabs)
Locust Workload Generator (300 Users, headless)